AREEBA AHMAD

AI Engineer | LLMs · RAG · Agentic AI · Computer Vision

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SUMMARY

AI Engineer (B.Sc. (Hons.) Information Technology, PUCIT, 2026) who builds LLM applications, RAG pipelines, multi-agent systems and computer vision models in Python. Built a legal-document RAG system with Weaviate at PixelPK Technologies, a 4-agent MCP/A2A system at PUCIT, and a YOLOv8 detector trained on 58,000+ images (85%+ mAP).

SKILLS

Languages: Python, SQL

GenAI & Agents: LLM APIs (Gemini, Groq), LangChain, RAG, prompt engineering, MCP, A2A, Google ADK, n8n

Vector Search: Weaviate, hybrid search, reranking, embeddings, HNSW

ML & Computer Vision: PyTorch, scikit-learn, YOLOv8, DINOv2, OpenCV

Backend & Cloud: FastAPI, REST APIs, asyncio, MySQL, AWS S3, Git

EXPERIENCE

AI/ML Intern, PixelPK Technologies (Remote) **Jun 2026 - Jul 2026**

- Built a pipeline that scrapes court judgments, converts PDFs to Markdown, extracts metadata with an LLM (Groq), stores files in AWS S3 and serves them through a FastAPI API.
- Developed a Weaviate RAG system with hybrid search and reranking for legal-judgment retrieval.
- Designed the scraper to be court-agnostic, so new courts can be added without reworking the core code.

AI Developer Intern, PUCIT, Lahore **Dec 2025 - May 2026**

- Architected a 4-agent system (Orchestrator, Generator, Validator, UI Controller) with FastAPI, asyncio and JSON-RPC 2.0; added per-agent telemetry with Google ADK.
- Built MCP servers (fastmcp) that let LLM agents call databases as tools.
- Built RAG workflows with Gemini 2.5 Flash and n8n on the WhatsApp Business API, serving 50+ concurrent sessions.

Web Development Intern, BixForgeSolutions (Remote) **Jun 2025 - Aug 2025**

- Sped up a CMS MySQL backend by fixing N+1 queries and adding missing indexes.

PROJECTS

MLOps Autopilot Auditor (MCPValidator)

- Clones a GitHub repository, scores its MLOps maturity from 0 to 100, uses Gemini 2.5 Flash to generate a Dockerfile, mlflow.yaml and tests, and opens a pull request with the fixes. *FastAPI, GitPython, PyGithub, n8n*

Ring Design Recommendation System

- Image-search API that finds similar jewelry designs using DINOv2 embeddings and an HNSW index; compared 2 fine-tuned models with Recall@K and mAP@K. *PyTorch, FastAPI*

AI-Based Real-Time Vehicle Smoke Detection and Reporting System (Final Year Project)

- Trained a YOLOv8 model on 58,000+ annotated smoke and fire images, reaching 85%+ mAP. *PyTorch, OpenCV, Roboflow*

EDUCATION

B.Sc. (Hons.) Information Technology, PUCIT, University of the Punjab **2022 - 2026**

Teaching Assistant, Software Engineering, PUCIT (Nov 2025 - Apr 2026)

CERTIFICATIONS

Google Generative AI Leader Specialization (2026) | Python for Everybody Specialization, University of Michigan (2026)